



8/7/2026

ASSIGNMENT DAY4

Transmission Media Analysis & RJ-45 UTP
Fabrication.

TASK 1.

❖ List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).

➤ Ethernet Cables vs Fiber Optic Cables :

3 Key Differences :

Working Method

- ⌘ Ethernet Cable: Transmits data using electrical signals.
- ⌘ Fiber Optic Cable: Transmits data using light signals.

Speed and Distance

- ⌘ Ethernet Cable: Suitable for short distances with good speed.
- ⌘ Fiber Optic Cable: Provides much higher speed and can transmit data over long distances with very low signal loss.

Cost

- ⌘ Ethernet Cable: Affordable and easy to install.
- ⌘ Fiber Optic Cable: More expensive but offers better performance and reliability.

TASK 2.

- ❖ Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.

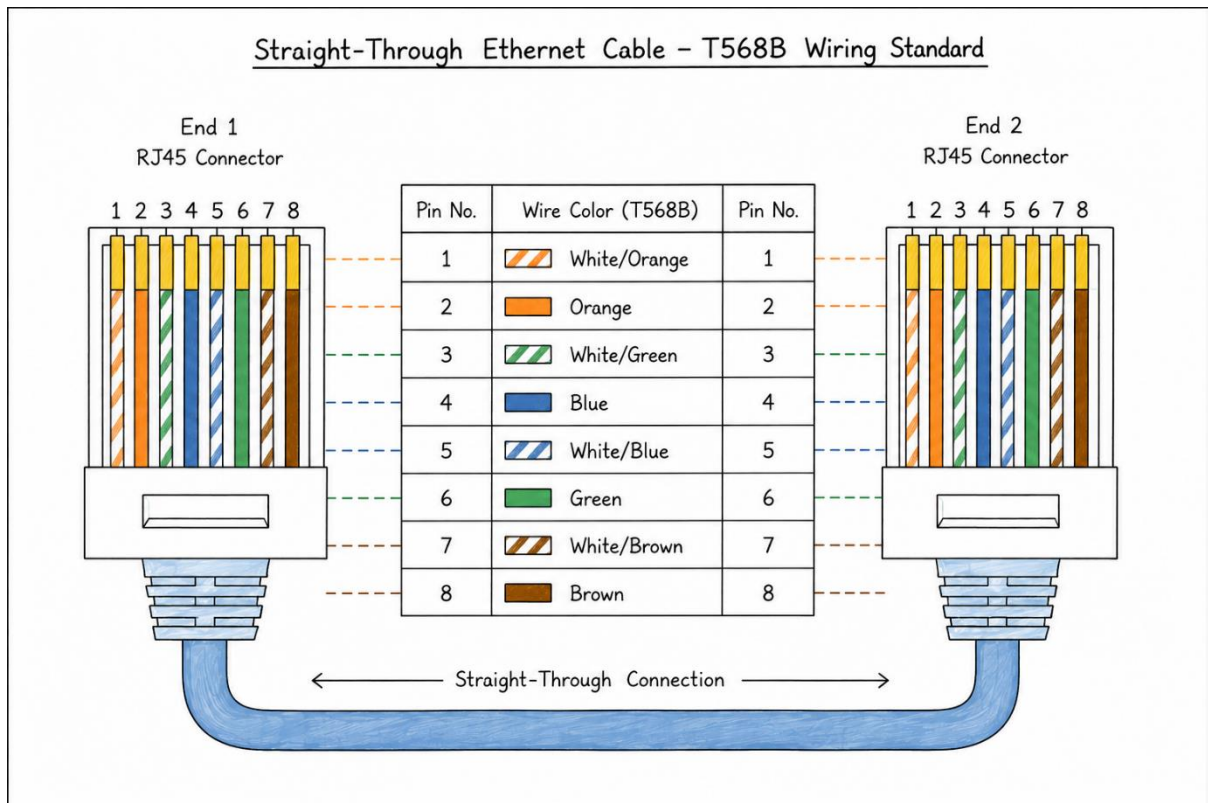


Figure : Internal Wiring of a Straight-Through Ethernet Cable (T568B Wiring Standard).

Explanation :

- ✓ The diagram shows the internal wiring of a straight-through Ethernet cable using the **T568B wiring standard**. Both RJ45 connectors follow the same wire sequence, with pins numbered from **1 to 8**. Each wire is labeled with its correct colour to ensure proper data transmission.

TASK 3.

❖ Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.

➤ Step-by-Step Guide to Attach an RJ45 Connector to a CAT6 Cable :

- ✂ Cut the CAT6 cable to the required length.
- ✂ Strip about 2–3 cm of the outer jacket to expose the eight wires.
- ✂ Untwist the wire pairs and arrange them in the T568B color order.
- ✂ Straighten the wires and trim them to the same length.
- ✂ Insert the wires into the RJ45 connector until they reach the end.
- ✂ Check that the wire order is correct and the cable jacket is inside the connector.
- ✂ Place the connector in the RJ45 crimper and press the handles firmly to crimp it.
- ✂ Test the cable using a cable tester to ensure it works properly.

TASK 4.

❖ Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences. Consider speed, distance, and budget for your answer.

➤ Cable and Tool Selection for a Gaming Cafe :

- ✓ For a gaming cafe with 10 PCs, I would use Ethernet cables and an RJ45 crimper. Ethernet provides fast and stable connections for LAN gaming over short distances. It is also more affordable than fiber optics, making it the best choice for a gaming café. An RJ45 crimper is used to attach RJ45 connectors to Ethernet cables, ensuring reliable network connections.